

# Vincent Griffo

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## EDUCATION

**Georgia Institute of Technology**, Atlanta, GA *Expected: December 2026*  
*Bachelor of Science in Mechanical Engineering; Minor in Physics; Concentration in Robotics* *GPA: 4.00/4.00*

**Harrison High School**, Harrison, NY *June 2022*  
*High School Diploma: Summa Cum Laude* *GPA: 4.00/4.00*

## AWARDS

**Tohoku Short-Term Trainee** *May 2026*  
Selected from an applicant pool by Tohoku University's School of Engineering to conduct research in Japan under a faculty member of my choice.

**Astronaut Scholarship** *May 2026*  
National scholarship awarded to top undergraduate researchers demonstrating exceptional academic achievement

**Goldwater Scholar** *March 2026*  
Competitive merit scholarship awarded to undergraduate students with strong research experience and academics

**Georgia Tech Capstone Honorable Mention** *April 2026*  
Awarded Honorable Mention for *Sidewalk Sidekick* during the Georgia Tech Spring 2026 Capstone Design Expo, recognizing one of the top 8 non-winning teams among more than 200 capstone teams university-wide.

**RSS Space Robotics Workshop Best Paper** *June 2025*  
Received Best Paper for GROVER (variable-height grouser wheel) at the RSS Space Robotics Workshop 2025

**ThinkSwiss Research Fellowship** *February 2025*  
Recipient of a competitive award to research at ETH Zurich under Dr. Marco Hutter of the Robotic Systems Lab

**President's Undergraduate Salary Award x2** *November 2024, November 2025*  
Awarded a competitive research grant to support research in the Aerospace Robotics Lab

**University Rover Challenge Final Round x2** *March 2024, March 2025*  
Rover advanced to the final round of the University Rover Challenge, competing against 32 international teams at the Mars Desert Research Station in Hanksville, Utah

**Stamps President's Scholar** *August 2022*  
Full-ride merit scholarship awarded to the top 1% of incoming students at Georgia Tech based on commitment to scholarship, leadership, progress, and service

## RESEARCH

**Aerospace Robotics Lab** | Georgia Tech *August 2024 – Present*  
*Principal Investigator: Prof. Yashwanth K. Nakka*

*Design, Control, and Testing of a Highly Dexterous Multimodal Rover*

- Lead a team of seven engineers designing an 8-DOF multimodal rover leg to experimentally study how leg dexterity and morphology influence locomotion efficiency and stability on lunar analog terrain
- Designed low-inertia serial leg architectures using inverse dynamics with MuJoCo and Pinocchio to evaluate joint torque requirements and actuator feasibility across multiple gait morphologies
- Integrated mechanical differentials and transmission strategies to reduce reflected inertia and peak joint torques during multimodal gait transitions

*Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals*

- Led a multidisciplinary team of six engineers while serving as the project's sole mechanical and controls engineer, developing a 12-DOF (3-DOF/leg) wheeled quadruped capable of 2.7 m/s rolling and 1.05 m/s walking
- Designed the complete mechanical system in SolidWorks, performing inverse dynamics in MuJoCo for actuator sizing and engineering high-load shafts, bearings, transmissions, and structural members validated through finite element analysis on seven machined aluminum components
- Developed the robot's control architecture using rigid-body dynamics, impedance control, dynamic gait generation, CAN bus communication, and EKF-based state estimation for robust multimodal locomotion

- Integrated a Jetson Nano autonomy stack implementing YOLO-based computer vision, obstacle detection, and A\* path planning for autonomous navigation and obstacle avoidance
- Manufactured and assembled custom components through CNC machining and waterjet fabrication, culminating in full-system integration and user testing with blind and low-vision participants

#### *Design of Rover Mobility Fleet for Multi-Agent Collaboration (NOVA)*

- Led the design of a low-cost collaborative rover fleet for multi-agent autonomy experiments across ARL's lunar terrain testbed to inform future lunar cobot missions
- Designed and manufactured the full rover mechatronic system, including chassis, rocker suspension, and four independent swerve-drive modules enabling omnidirectional motion at speeds up to 2 m/s
- Architected the complete 12 V electrical system, including battery specification, power distribution, actuator selection, relays, and dual Jetson Nano integration
- Integrated dual ZED Mini stereo cameras and onboard perception to enable cooperative navigation and collaborative environment understanding across multiple rovers

#### *Design and Control of Variable-Height Grouser Platform for Extraterrestrial Navigation*

- Invented and designed an internal spiral cam mechanism within a wheel that dynamically adjusts grouser height to match analog Martian terrain properties, achieving up to a 58% reduction in slip ratio in granular media and informing how adaptability may aid crucial locomotion in future Martian missions
- Performed structural FEA in SOLIDWORKS on spiral cam and tread geometries, informing stiffness, contact stress, and fatigue limits under dynamic terrain interaction loads
- Designed a fully instrumented terrain testing gantry with linear and rotary encoders, current and voltage sensing to experimentally quantify slip ratio, tractive force, and energy consumption across four terrains
- Developed a closed-loop PID controller in Python to precisely regulate grouser geometry, enabling accurate and repeatable experimental validation of wheel-terrain interaction

### **Space Infrastructure Engineering Lab | Tohoku University, Japan**

*May 2026 – July 2026*

*Principal Investigator: Prof. Toshinori Kuwahara*

#### *Design and Control of Electromechanical Soft Docking System for Floating Platform Cubesat*

- Designed and integrated a mountable soft-capture docking mechanism with complementary male and female interfaces for autonomous CubeSat rendezvous, capture, and mechanical locking.
- Designed a servo-actuated twist-lock mechanism with ball-detent alignment features, enabling low-friction docking and repeatable locking and undocking operations.
- Designed, wound, and optimized a custom iron-core electromagnet from first principles to enable long-range electromagnetic soft capture on the Space Infrastructure Engineering Lab's air-bearing test platform.
- Applied electromagnetic first principles and a current-controlled test system to maximize magnetic flux while minimizing thermal losses, achieving reliable capture distances of up to 37 cm.

### **Robotic Systems Lab / LunarLeaper | ETH Zurich, Switzerland**

*August 2025 – December 2025*

*Principal Investigator: Prof. Marco Hutter*

#### *Design and Control of a 3-DOF Parallel, Space-Grade Arachnid Robot (LunarLeaper)*

- Led mission-driven mechanical design of a 3-DOF parallel leg mechanism for *LunarLeaper*, a 12U CubeSat-class quadrupedal robot planned for a 2030 lunar science exploration mission
- Iterated through six full-scale leg prototypes using rapid prototyping and COTS hardware to converge on a compact architecture capable of surviving lunar dust, thermal cycling, and launch constraints
- Executed full mechatronic development of the final prototype using PMSM actuators, employing stationary actuator placement to eliminate moving thermal straps and wiring, improving system lifetime
- Leveraged joint torque and ROM analysis in Simscape Multibody to implement final transmission architectures using heim-joint yaw and hip pitch joints and a steel-shaft flexure knee mechanism

### **Aerial Robotics and Experimental Autonomy Lab | Georgia Tech**

*August 2023 – August 2024*

*Principal Investigator: Prof. Jonathan Rogers*

### *Design and Control of Multimodal UAS Proprotor Articulation for VTOL-to-Fixed-Wing Transition*

- Worked on a novel proprotor articulation mechanism to allow seamless transition between vertical take-off and landing (VTOL) and fixed-wing flight modes for a scaled-down unmanned aerial vehicle (UAV)
- Created mathematical models and performed simulations of articulation using MATLAB, applying principles of rigid body mechanics to optimize spatial constraints, maximize torque output, and enhance moment arm rotation
- Designed and implemented a PID control system in MATLAB Simulink, integrating hardware-in-the-loop (HIL) simulation with a DC motor and Arduino Nano for application in the UAS control scheme

**Laboratory for Intelligent Decision and Autonomous Robots** | Georgia Tech     *December 2022 – March 2023*  
*Principal Investigator: Prof. Ye Zhao*

### *Development of Control Systems and Autonomous Navigation for Human-Avoidance Drones*

- Leveraged ROS and Python to design autonomous human-avoidance algorithms to enable real-time drone navigation and obstacle detection for enhanced flight path control

**SSP Astrophysics** | Online through MIT, Caltech, and Harvey Mudd Sponsorship     *May 2021 – July 2021*  
*Principal Investigators: Prof. Michael Hannawald & Prof. Cassandra Fallscheer*

### *Tracking, Modeling, and Orbital Prediction of Near-Earth Asteroid 2003 HB6*

- Determined the position of an asteroid relative to known stars using parallax and coordinate transformations with data from industry-grade telescopes at observatories in Chile, Australia, and the USA
- Calculated orbital parameters of asteroids utilizing the Method of Gauss and Monte Carlo simulation to be used for a precise visual model and accurate approximations within 0.1% of JPL's previously predicted orbital elements
- Developed a mathematical model and 3D visual simulation using Python to predict asteroid positions decades in advance based on determined orbital parameters
- Collaborated with the Southwest Research Institute to analyze asteroid collision history using advanced orbital simulations developed with SWIFT

## **JOURNAL PUBLICATIONS**

- [1] **V. Griffo**, Y. Nakka, Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation (**Submitted IEEE T-FR, revisions requested**)
- [2] **V. Griffo**, B. Ma, J. Yu, Data on Orbit Determination of Asteroid 326732 (2003 HB6) (**Internal to the Harvard Minor Planet Center**)

## **CONFERENCE PUBLICATIONS**

- [1] Y. Nakka, **V. Griffo**, S. Kumar, C. DelGaudio, T. Dokania, A. Bhardwaj, S. Kelkar, U.K. Ponugupati. A Lunar Terrain Facility for Robust and Resilient Learning-Based Autonomy. *IEEE Aerospace Conference, March 2027* (**Submitted**)
- [2] J. Song **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium October 2026*
- [3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE, September 2026*
- [4] S. Stähler, A. Mittelholz, A. Cocheril, A. Fuhrer, J. Church, E. Krasnova, H. Kolvenbach, J. Smajic, V. Bickel, M. Betschen, E. Brändli, J. Gomez Jodar, **V. Griffo**, L. Heyun, N. Hirzel, Y. Hofmann, J. Isler, Y. Luginbühl, J. Majer, R. Margarit, G. Smith-Vaniz, M. Hutter, R. Kornfeld, LunarLeaper: A university-led legged robotic mission to the moon. *5th Symposium on Space Educational Activities April 2026*.
- [5] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration. *Lunar Surface Innovation Consortium November 2025* (**Accepted, Canceled due to Government Shutdown**)
- [6] S. Kumar, **V. Griffo**, Y. Nakka, NOVA: Design Overview of an Autonomous 3D-printed Rover Platform. *Lunar Surface Innovation Consortium, November 2025* (**Accepted, Canceled due to Government Shutdown**)
- [7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics, June 2025* (**Best Workshop Paper**)

## ORAL PRESENTATIONS

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- [1] **V. Griffo**, T. Kuwahara, Design and Control of an Electromechanical Soft-Docking System for Floating-Platform CubeSats. *Tohoku University LQS*, Sendai, Japan, July 2026.
- [2] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.
- [3] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration *Lunar Surface Innovation Consortium*, November 2025 (**Accepted, Canceled due to Government Shutdown**)
- [4] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, June 2025
- [5] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech Annual Research Symposium*, April 2025

## POSTER PRESENTATIONS

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- [1] J. Song **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium* October 2026
- [2] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE*, September 2026
- [3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *GT Space Week*, September 2026
- [4] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.
- [5] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Lunar Surface Innovation Consortium*, November 2025 (**Accepted, Canceled due to Government Shutdown**)
- [6] S. Kumar, **V. Griffo**, Y. Nakka, NOVA: Design Overview of an Autonomous 3D-printed Rover Platform. *Lunar Surface Innovation Consortium*, November 2025 (**Accepted, Canceled due to Government Shutdown**)
- [7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, June 2025
- [8] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech College of Sciences EXPLORE*, March 2025
- [9] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Stamps Scholar Summit*, March 2025

## PATENTS

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- [1] **V. Griffo**, Adaptive Grouser Wheel for Planetary Rovers. *US Provisional Patent Application* No. 63/903,602, filed 22 October 2025. GTRC Ref. 2026-079 (GTRC2026079PRV).

## PROFESSIONAL EXPERIENCE

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**Boston Dynamics**, Waltham, MA

July 2026 – Present

*ATLAS Systems Test / Mechanical Engineering Co-Op*

- Designing and developing custom test rigs to evaluate the dexterous manipulation capabilities of Atlas' end effectors, validating tasks including lifting, reaching, and fine tool manipulation.
- Performing structural analysis and finite element analysis (FEA) to ensure test fixtures exceed required load and stiffness margins, preventing the test apparatus from becoming the failure mode during evaluation.
- Integrating load cells, force/torque sensors, data acquisition systems, and signal conditioning electronics to quantify gripper performance across representative manipulation tasks.

- Designing and validating leg test fixtures to characterize Atlas' lower-limb performance throughout multiple stages of the walking lifecycle using principles of legged robot dynamics.
- Developing mechanical designs, engineering drawings, and production-ready documentation using CATIA V6 while collaborating with multidisciplinary hardware, controls, and test engineering teams.

**MIT Lincoln Laboratory**, Lexington, MA

*May 2024 – August 2024*

*Interceptor Technology and Systems Architecture Researcher*

- Designed and developed a low-cost counter-UAV interceptor, reducing manufacturing time and costs via 3D printing and fastener-free design, set for implementation in operational missions for rapid threat response
- Designed, prototyped, and flight-tested a custom 3D-printed high-torque steering control assembly for a low-cost counter-unmanned aerial system (C-UAS) interceptor missile, enhancing maneuverability and control precision
- Designed and manufactured a deployable fin system utilizing quaternions and eigenvalue decomposition, achieving a 60% reduction in the stowed rocket volume, allowing compact stacking and launching of the interceptors
- Helped develop key subsystems, including PCB avionics, and integrated YOLO-based computer vision for real-time target detection, enhancing autonomous threat response for operational deployment
- Implemented physics-based models using MATLAB and MODTRAN to conduct trade studies of electro-optical/infrared (EO/IR) sensors for unmanned vehicles under adverse atmospheric conditions

**Georgia Tech Undergraduate Teaching Assistant**, Atlanta, GA

*Fall 2023, Fall 2024*

*Department of Mathematics Linear Algebra Teaching Assistant*

- Independently instructed over 40 students in linear algebra, developing comprehensive lecture materials focused on deepening student understanding of key concepts such as eigenvalues, matrix transformations, and vector spaces
- Collaborated with a team of 20+ TAs and professors to coordinate structured grading rubrics and ensure consistent instruction through clear communication and alignment on course objectives and evaluation criteria

## COMPETITION TEAMS

**URC Mars Rover Team** | *Chief Engineer*

*September 2022 – Present*

- Oversee the design of an autonomous rover consisting of a rocker-bogie drivetrain, 5-DOF robotic arm, precision end-effector, and carbon fiber chassis for simulated Martian navigation in the University Rover Challenge
- Designing a highly dexterous serial 5-DOF robotic arm using Simscape Multibody analysis, validating rigid-body dynamics and torque requirements for manipulation of a 5 kg payload at 1 m reach
- Designed and fabricated a compliant end-effector to manipulate objects with uncertain geometries, improving operational dexterity across competition tasks
- Led the design and manufacture of a custom carbon fiber drone frame and control scheme, featuring a PID-controlled camera gimbal system integrated with a CUAV X7 flight controller for stabilization and flight control
- Designed and manufactured a 2-DOF robotic arm for the URC drone, enabling manipulation and transport of hand tools for mission-critical operations

**Georgia Tech Robotics Competition**

*August 2023 – November 2023*

- Competed against over 200 mechanical engineering students in the Georgia Tech robotics competition, designing and programming an autonomous robot within 5 weeks
- Led the development of a telescoping lift, custom gear drivetrain, high-traction SLA tires, and a passive geometry ball filter, ultimately achieving first seed before the final competition out of 67 teams

## VOLUNTEER WORK

**Invention Studio Makerspace** | *Electronics Apprentice*

*September 2022 – March 2024*

- Instructed and guided students and makers in the use of various machining tools and techniques, supporting both research and personal projects within the makerspace
- Demonstrated proficiency in circuit design, electromagnetism, microcontrollers, and shop management by creating complex electronic projects, such as a Gauss Accelerator, and managing the budget to order and restock makerspace components

## Skills

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- **Technical Software:** SOLIDWORKS, SOLIDWORKS FEA, Siemens NX, Fusion 360, ANSYS, MATLAB, Simulink, Simscape Multibody, MuJoCo, Mission Planner, PX4, Jetson SDK, Linux
- **Hardware & Prototyping:** Precision machining (lathe, mill, waterjet, PCB mill), additive manufacturing (FDM/SLA), TIG/MIG welding, soldering and wiring, sensor integration, rapid prototyping, electromechanical assembly
- **Programming & Embedded Systems:** C++, Python, MATLAB, C#, Java, JavaScript, ROS, Arduino IDE, Raspberry Pi OS,  $\text{\LaTeX}$
- **Robotics & Control:** Robot kinematics and inverse dynamics, Jacobian-based analysis, legged locomotion, state estimation, actuator design and sizing, PID and field-oriented control (FOC), embedded serial interfaces, hardware-in-the-loop (HIL) control, data acquisition